A proof of the Kauers–Koutschan conjecture for OEIS A195806

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Abstract

We prove Conjecture 11 of Kauers and Koutschan (*Some D-finite and some possibly D-finite sequences in the OEIS*, arXiv:2303.02793, J. Integer Seq. 26 (2023), 23.4.5): the sequence OEIS A195806 is given by an explicit quasi-polynomial of degree 6 and period 6, and consequently satisfies the conjectured order-14 constant-coefficient linear recurrence. The proof identifies the sequence as the Ehrhart counting function of an explicit rational polytope of dimension 6 and denominator 6, which reduces the conjecture to a finite, exact computation.

1 The sequence and the conjecture

OEIS A195806 (R. H. Hardin, 2011) counts *triangular* $5 \times 5 \times 5$ 0..*n* *arrays with all rows and diagonals of the same length having the same sum, with corners zero.* Precisely (the reading is calibrated against the published data in Section 2): consider a triangular array $x = (x_{i,j})_{0 \le j \le i \le 4}$ of 15 integers. Its *lines* are the fifteen maximal segments in the three lattice directions:

$$\text{rows } \{x_{i,j} : 0 \le j \le i\} \ (i \text{ fixed}), \qquad \text{left diagonals } \{x_{i,j} : j \le i \le 4\} \ (j \text{ fixed}),$$
$$\text{right diagonals } \{x_{i,i-k} : k \le i \le 4\} \ (k \text{ fixed}),$$

three lines of each length $\ell \in \{1, \dots, 5\}$. The array is *admissible* if any two lines of equal length have equal sums, and the three corner entries $x_{0,0}, x_{4,0}, x_{4,4}$ are zero. Then

$$a(n) \ = \ \#\{\, x \text{ admissible} : 0 \le x_{i,j} \le n \text{ for all } i, j \,\}.$$

Theorem 1 (= Kauers–Koutschan Conjecture 11)**.** *For all $n \ge 0$, writing $n = 6m + r$ with $r \in \{0, \dots, 5\}$,*

$$a(n) \ = \ Q_r(m),$$

where

$$Q_0(m) = 1 + 25m + 158m^2 + 650m^3 + 2275m^4 + 4680m^5 + 4680m^6,$$
$$Q_1(m) = 16 + 198m + 1133m^2 + 3900m^3 + 8125m^4 + 9360m^5 + 4680m^6,$$
$$Q_2(m) = 105 + 1087m + 4922m^2 + 12350m^3 + 17875m^4 + 14040m^5 + 4680m^6,$$
$$Q_3(m) = 496 + 4148m + 14783m^2 + 28600m^3 + 31525m^4 + 18720m^5 + 4680m^6,$$
$$Q_4(m) = 1759 + 12121m + 35258m^2 + 55250m^3 + 49075m^4 + 23400m^5 + 4680m^6,$$
$$Q_5(m) = (1 + m)^2\,(5052 + 19370m + 28405m^2 + 18720m^3 + 4680m^4).$$

[*]Produced as part of a two-agent collaboration set up by Jonathan Bostock, 2026-07-22. All computations are reproducible from the scripts in `work/claude/a195806_*.py` of the accompanying repository; Codex's independent verification is `work/codex/review_a195806.py`.

Consequently a satisfies, for all $n \geq 0$, the recurrence

$$a(n) - 3a(n{+}1) + 2a(n{+}2) - a(n{+}3) + 6a(n{+}4) - 5a(n{+}5) - 3a(n{+}6) \\ + 3a(n{+}8) + 5a(n{+}9) - 6a(n{+}10) + a(n{+}11) - 2a(n{+}12) + 3a(n{+}13) - a(n{+}14) = 0.$$

2 Calibration of the definition

The OEIS name is terse, so we first fix the reading above against the published data. A direct brute-force count over all $(n{+}1)^{12}$ assignments of the twelve non-corner cells yields $a(1) = 16$, $a(2) = 105$, $a(3) = 496$, in agreement with the OEIS entry; moreover the exact counting procedure of Section 5 reproduces *all* 32 terms of Hardin's b-file (`b195806.txt`, archived with the repository). We are therefore certainly counting the objects of A195806.

3 The integer solution lattice

Let $A \in \mathbb{Z}^{13\times 15}$ be the constraint matrix consisting of ten equal-sum equations (for each length ℓ, two independent differences among the three line sums) and three corner equations $x_{0,0} = x_{4,0} = x_{4,4} = 0$. Exact computation gives $\operatorname{rank} A = 9$.

Lemma 2. *There is a matrix $K \in \mathbb{Z}^{15\times 6}$ whose columns form a basis of the full integer kernel* $\ker A \cap \mathbb{Z}^{15}$*; i.e. $z \mapsto Kz$ is a bijection $\mathbb{Z}^6 \to \{x \in \mathbb{Z}^{15} : Ax = 0\}$.*

Proof. Compute a Smith normal form $UAV = S$ with U, V unimodular (verified exactly in rational arithmetic in the accompanying script). Exactly 9 invariant factors are nonzero, so for $y \in \mathbb{Z}^{15}$, $A(Vy) = 0$ iff the first 9 coordinates of y vanish; since V is a bijection of $\mathbb{Z}^{15}$, the last 6 columns of V form the desired basis K.

Independently (Codex's certificate): $AK = 0$ and $\operatorname{rank} K = 6$ are verified directly, and the 6×6 submatrix of K on rows $(2, 8, 9, 11, 12, 13)$ (0-based) has determinant -1. A full-column-rank integer matrix with a unimodular 6×6 minor spans a *primitive* sublattice (the minor shows every rational vector in the column span with integer image under those six coordinates is an integer combination), so the column lattice of K equals $\ker A \cap \mathbb{Z}^{15}$. □

Three rows of K are zero (the corners) and five rows are the unit vectors e_1, e_3, e_4, e_5, e_6; the full matrix is printed by the script and archived. By Lemma 2,

$$a(n) \;=\; \#\{\, z \in \mathbb{Z}^6 \;:\; 0 \leq (Kz)_i \leq n,\; 1 \leq i \leq 15 \,\}. \tag{1}$$

4 Ehrhart structure

Let

$$P \;=\; \{\, z \in \mathbb{R}^6 : 0 \leq (Kz)_i \leq 1 \text{ for all } i \,\},$$

so that by (1), $a(n) = \#(nP \cap \mathbb{Z}^6) = L_P(n)$ is the Ehrhart counting function of P.

Proposition 3. *P is a bounded rational polytope of dimension 6 with 58 vertices, and every vertex coordinate has denominator dividing 6.*

Proof. K has rank 6, so $z \mapsto Kz$ is injective and P, the preimage of the box $[0,1]^{15}$, is bounded. Since P is a bounded polyhedron defined by the 24 nontrivial inequalities (twelve nonzero rows of K, two bounds each), every vertex is the unique solution of six linearly independent tight constraints. Enumerating all $\binom{24}{6} = 134596$ subsets in exact rational arithmetic and retaining the feasible solutions yields 58 vertices; their affine hull has dimension 6; the least common multiple of all coordinate denominators is 6. This computation was performed independently twice (Claude and Codex) with matching results. □

Proposition 4 (Ehrhart)**.** *$a(n)$ is a quasi-polynomial in n of degree 6 whose period divides 6, valid for all integers $n \geq 0$.*

Proof. By Ehrhart's theorem for rational polytopes (Ehrhart 1962; see e.g. Stanley, *Enumerative Combinatorics* I, §4.6, or Beck–Robins, *Computing the Continuous Discretely*, Thm. 3.23), L_P is a quasi-polynomial of degree $\dim P = 6$ whose period divides the denominator of P (the lcm of the vertex coordinate denominators), here 6, by Proposition 3. □

5 Finite verification

Both $a(n)$ (by Proposition 4) and the conjectured Q (by inspection) are quasi-polynomials of period 6 and degree ≤ 6. Fix $r \in \{0, \ldots, 5\}$: then $m \mapsto a(6m+r)$ and $m \mapsto Q_r(m)$ are polynomials of degree ≤ 6, hence equal identically as soon as they agree at seven values of m. It therefore suffices to check $a(n) = Q(n)$ for $0 \leq n \leq 41$.

The values $a(0), \ldots, a(41)$ are computed exactly from (1) by integer enumeration: five coordinates of z range over $[0, n]$ (the unit rows of K), the sixth is counted by intersecting the integer intervals imposed by the remaining rows; the hard-coded linear forms are machine-checked against K at import time. The computation (a few seconds) gives exact agreement with Q at all 42 points; as noted in Section 2 it also reproduces the 32 b-file terms. Codex independently recomputed $a(0..12)$ with a generic counter derived mechanically from K, and re-verified the seven interpolation points $n = 35, \ldots, 41$. This proves the quasi-polynomial claim of Theorem 1.

Finally, the recurrence: its left-hand side, applied to Q, is again a quasi-polynomial of period 6 and degree ≤ 6 in n, so its vanishing for all n follows from its (verified) vanishing at $n = 0, \ldots, 47$. □

Remark 5. $a(0) = 1 = Q_0(0)$ (the all-zero array), consistent with extending the offset-1 OEIS sequence to $n = 0$; the interpolation argument uses $n \geq 0$, comfortably inside Ehrhart validity.

Remark 6 (What is computer-verified). Rank and Smith normal form of A (exact, with the unimodular-minor certificate as an independent check); the vertex enumeration (twice, independently); the 42 exact counts (twice, independently, plus the 32-term b-file cross-check); the recurrence evaluation. No floating point is used anywhere.